

SILA BOZKURT

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SUMMARY

Dedicated second-year Computer Science and Engineering student passionate about machine learning, cloud computing, and robotics. Seeking internship opportunities to apply strong programming and data-driven problem-solving skills to real-world tech challenges and innovative solutions.

EDUCATION

Bilkent University

B.Sc. in Computer Science & Engineering

Sept 2024 – June 2028 (Expected)

Honor's Certificate (2024-2025)

Relevant Coursework: Algorithms and Programming, Linear Algebra and Differential Equations, Probability and Statistics for Engineers, Introduction to Macroeconomics, Data Structures, Digital Design and Computer Architecture, Computer Organization

EXPERIENCE

Bilkent Space and Aeronautics Society (BİLSAS)

Ankara, TR

Treasurer

Jan 2026 – Present

- **Generated** monthly financial audit reports for the executive board, ensuring fiscal transparency and 100% compliance with society regulations

Board Member & Member of the Great Assembly

Sept 2025 – Jan 2026

- **Directed** the approval workflow for society activities by conducting feasibility studies, leading to the successful execution of high-priority missions

BİLSAT Model Satellite Team

Ankara, TR

Leader of Ground Control Software Branch

Jan 2026-Present

- Architected real-time Ground Station GUI for a model satellite, ensuring zero latency while parsing 1 Hz live telemetry data.
- Designed an autonomous Interface Alarm System (ARAS) for flight tracking, capable of detecting payload separation anomalies and GPS failures within milliseconds, by developing custom fault-tolerance logic and telemetry parsing protocols.

PROJECTS

Ghost Robot – ROS-Based Autonomous Navigation

Feb 2026 – Present

- **Developing** an autonomous navigation stack using ROS and NVIDIA Isaac Sim, integrating LiDAR-based SLAM for real-time mapping
- **Designing** high-fidelity robot kinematics and control loops in C++ to achieve precise trajectory tracking in dynamic simulation environments
- **Implementing** multi-sensor fusion algorithms to process IMU and odometry data, reducing positional drift by optimizing EKF parameters

SeismIQ – Cloud-Native Disaster Response Platform | Team Founder

June 2025 – September 2025

- **Won 3rd Place** in the AWS Student Engagement Program by architecting an event-driven system using Lambda, SageMaker, and DynamoDB
- Architected end-to-end emergency notification system where earthquake events and user reports trigger AWS Lambda functions that process and route alerts through SNS topics to deliver real-time push notifications to Android devices based on location proximity and user roles, so that nearby users (calculated 10km radius based on the coordinates) were instantly alerted, enabling faster coordination and response
- **Visualized** real-time crisis data by integrating Google Maps SDK into a Java-based Android client, enhancing situational awareness

Autonomous UAV Hexacopter | Autonomous Flight Developer BİLSAS Team

Feb 2025 – Aug 2025

- Advanced as a finalist in the Teknofest UAV competition, ranking in the top 5% (out of 1037 teams)
- In the context of autonomous UAV missions, implemented real-time object detection with **OpenCV**, **cv2** with **other computer vision frameworks** and integrated **DroneKit** and **MAVLink** for flight and servo control on a Raspberry Pi + Navio2, enabling reliable perception-to-action for competition payload drops
- To improve navigation accuracy, modeled pixel-to-meter error considering altitude, latency, and sensor noise, achieving 0.15–0.3 m ground precision at 8-20 m altitude.
- **Stabilized** target engagement by combined geometric projection, adaptive pixel tolerance, and filtering techniques (EMA + hysteresis), reducing false triggers by over 40% during flight

Bio-Oracle – Computer Vision for Cell Tracking | Team Founder

Jan 2026 – Feb 2026

- **Designed** an automated cell tracking system through developing **my own framework** in which I got assistance from YOLO11 and PyTorch, utilizing image segmentation and trajectory analysis for microscopy datasets (EMDS-5 and 6)
- **Architected** a data extraction module in Python to model cell movement, achieving high accuracy in biological simulation environments

SKILLS

Python, Java, C++, C, SystemVerilog, MIPS, SQL (MySQL, PostgreSQL), scikit-learn, pandas, NumPy, matplotlib, Jupyter Notebook, Spring Boot, REST APIs, Android Studio (Java), Blynk, Arduino, Git/GitHub, AWS (Lambda, DynamoDB, S3, API Gateway, SageMaker), ROS, Isaac Sim, OpenCV, Linux, MAVLink, DroneKit French (A2), Russian(B1), German(B1)